

Amir Hossein Saleknia

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Education

M.S. IN DIGITAL ELECTRONIC SYSTEMS

Sep. 2019 - Sep. 2022

Iran University of Science and Technology (IUST)

Tehran, Iran

- **Thesis topic:** Development of a method for image segmentation based on knowledge distillation (Medical Images)
- **Supervisor:** Dr. Ahmad Ayatollahi, Full Professor at IUST
- **GPA:** 18.22/20 (4/4)

B.S. IN ELECTRICAL ENGINEERING

Sep. 2015 - Jul. 2019

University of Zanjan

Zanjan, Iran

- **Thesis topic:** Design Fractional-order controller for DC/DC convertors
- **Supervisor:** Dr. Abolfazl Jalilvand, Full Professor at University of Zanjan
- **GPA:** 18.30/20 (3.78/4)

Languages

English IELTS (Academic): 7.5, C1 Proficiency

Persian Native

Research Interests

- Computer Vision
- Deep Learning
- Knowledge Distillation
- Medical Image Analysis
- Pattern Recognition
- Image Segmentation
- Ensemble Learning
- Transformers

Publications

JOURNAL

- **A Comprehensive Study on Traffic Accident Detection, Contributing Factors, Analysis Methods, and Crash Datasets**
Elmira Bagheri, Amir Hossein Barshooi, **Amir Hossein Saleknia**
WILEY-HINDAWI, Journal of Advanced Transportation (JAT), 2024. **(Revised)**

CONFERENCE

- **Multi-source Ensemble Model for Scene Recognition**
Amir Hossein Saleknia, Ahmad Ayatollahi
IEEE, 14th International Conference on Computer and Knowledge Engineering (ICCKE), 2024. **(Revised)**
- **Multi Step Knowledge Distillation Framework For Action Recognition in Still Images**
Amir Hossein Saleknia, Ahmad Ayatollahi
IEEE, 20th CSI International Symposium on Artificial Intelligence and Signal Processing (AISP), 2024. **(Paper)**
- **NEM: Nested Ensemble Model for scene recognition**
Amir Hossein Saleknia, Elmira Bagheri, Amir Hossein Barshooi, Ahmad Ayatollahi
IEEE, 13th Iranian/3rd International Machine Vision and Image Processing Conference (MVIP), 2024. **(Paper)**
- **Efficient still image action recognition by the combination of ensemble learning and knowledge distillation**
Amir Hossein Saleknia, Ahmad Ayatollahi
IEEE, 9th International Conference on Web Research (ICWR), 2023. **(Paper)**

IN PROGRESS

- **MM-UNet: multi-encoder and multi-decoder network for medical image segmentation**
Amir Hossein Saleknia, Ahmad Ayatollahi
A manuscript is being prepared for submission to the Journal of Computers in Biology and Medicine (Elsevier).

Skills

Programming Languages	Python, Matlab, C, VHDL, LaTeX, Familiar with HTML, CSS, JavaScript
AI Frameworks & Libraries	PyTorch, TensorFlow, OpenCV, NumPy, Pandas, Matplotlib, etc.
Platforms & Tools	Jupyter Notebook, Google Colab, Kaggle, GitHub, etc.

Honors & Awards

- **Ranked 1st** among 40 peer B.Sc. students who chose Control as a subfield at University of Zanjan.
- **Ranked 3th** among 30 peer M.Sc. students of Digital Electronic Systems at Iran University of Science and Technology.
- **Ranked within the top 3%** among approximately 180,000 participants in the National-Wide University Entrance Exam in the Field of Physics and Mathematics for the B.Sc. degree.
- **Ranked within the top 1%** among approximately 15,000 participants in the National-Wide University Entrance Exam in the Field of Electronics for the M.Sc. degree.

Projects

- Design a novel network structure for skin lesion segmentation using Deformable attention and Multiscale feature aggregation.
- Extensive experiments with action recognition and scene recognition datasets to recognize their unique characteristics.
- Investigating the effectiveness of different knowledge distillation strategies for semantic image segmentation.
- Design and Analysis of fractional-order controller for DC/DC convertors based on Bee Colony optimization algorithm in Matlab.
- Using VHDL to implement various data encryption algorithms, including AES and DES.

Teaching Experience

2019	Linear Algebra , Teacher Assistant	<i>Zanjan, Iran</i>
2018	Electronic II , Teacher Assistant	<i>Zanjan, Iran</i>